

J.A.H.R. Weerasinghe

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Email

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Portfolio

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Professional Summary

Electronic and Telecommunication Engineering undergraduate with hands on expertise in embedded systems, digital signal processing, and robotics. Experienced in leading cross-functional teams for national hardware competitions, with proficiency in C/C++, CAD design, and PCB design.

Experience

Embedded Systems Developer

Designing and developing IoT-enabled smart devices and data acquisition systems. Experienced in C/C++ firmware development for STM32 and ESP32 microcontrollers, focusing on peripheral configuration (DMA, Timers, Interrupts), sensor integration, and custom PCB design.

Robotics System Developer

Engineering autonomous robotic platforms with a focus on high-frequency, real-time PID control and dynamic navigation. Prototyping mechanical hardware and writing optimized motor control firmware utilizing STM32 and Arduino architectures.

Tutor

Tutoring local and international (Edexcel) students in Advanced Level Mathematics and Physics.

Education

University of Moratuwa

2024–Present

BSc (Hons) in Electronics and Telecommunications Engineering

CGPA: 3.80 / 4.00

Kegalu Vidyalaya

2014–2022

G.C.E. Advanced Level (2022/23): Z-Score 2.7911 (Island 49th, District 2nd)

G.C.E. Ordinary Level (2019): 9 'A' Passes

Technical Skills

- **Programming:** Python, C/C++, MATLAB, Java
- **PCB Design:** Altium
- **CAD Design:** SolidWorks
- **Graphic Design:** Blender
- **Microcontrollers:** STM32, ESP32, Arduino

Projects

• Single Phase Power Quality Analyzer (2026)

[View Project](#)

- Measured and analyzed key electrical parameters—including RMS, total power, and harmonics up to the 25th order—at a 10 kHz sampling rate.
- Visualized calculated parameters in real-time via an integrated LCD and a web-based dashboard.
- *Individual contribution:* Designed the Analog-to-Digital Conversion (ADC) circuitry and the Digital Signal Processing (DSP) calculation architecture, implemented natively on an STM32 microcontroller.
- *Tools:* C/C++ (ESP32), STM32, Python

• Micromouse (Autonomous Maze-Solving Robot) (2025–2026)

[2026](#) | [2025](#)

- Engineered an autonomous robot capable of mapping and navigating unknown maze configurations under tight time constraints.
- Developed a high-frequency (1 kHz) PID motor control loop and implemented the Floodfill algorithm with optimized trajectory planning for smooth, multi-cell cornering.
- *Tools:* C/C++, STM32CubeIDE, STM32CubeMX, SolidWorks, Altium Designer

• 10 Band Analog Audio Frequency Spectrum Analyzer (2025)

 [View Project](#)

- Developed a fully analog spectrum analyzer to split and visualize audio input across 10 distinct frequency bands.
- *Individual contribution:* Designed the 10-band frequency splitter circuit, conducted LTspice simulations to verify frequency response, and optimized component selection.
- *Tools:* LTspice, SolidWorks, Altium Designer

• IoT Timer Smart Wall Plug (2025)

 [View Project](#)

- Built an ESP32 based IoT-enabled wall adapter featuring a built-in timer with support for both remote (online via mobile app) and physical (offline) scheduling.
- *Individual contribution:* Designed the custom PCB and developed the firmware for the ESP32 micro-controller prototype.
- *Tools:* C/C++ (Arduino IDE), ESP32, SolidWorks, Altium Designer

• Smart Cattle Collar (Ongoing)

- Developing an IoT-enabled wearable collar for cattle to monitor environmental methane emissions and track GPS location.
- Integrated a machine learning model utilizing microphone and accelerometer data for estrus (heat) detection.
- *Individual contribution:* Led the hardware development, including circuit design and multi-layer PCB layout.
- *Tools:* C/C++ (PlatformIO), ESP32, Python (Raspberry Pi), SolidWorks, Altium Designer

• Autonomous Robot for SLRC 2026

 [View Project](#)

- Built an autonomous mobile robot for the SLRC 2026 competition capable of line following, wall following, maze navigation, box detection, and automated box pickup.
- Implemented sensor-based navigation enabling seamless mode-switching between line following and wall following, along with autonomous maze solving.
- *Individual contribution:* Designed the motion control algorithms for line following, wall following, straight-line driving, and precision turns.
- *Tools:* C/C++ (Arduino IDE), YOLO model deployed on Raspberry Pi 4 for box detection, SolidWorks

Competitions

- RoboRoarz Champions (2026)
- Sri Lanka Robotics Challenge/SLRC Finalist (2026)
- SPARK Challenge Finalist (2026)
- Robofest Micromouse Finalist (2025)